

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 15: Light

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	15 — Light
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. the hardest traps are multi-step image-character chains, lateral-inversion of asymmetric letters/clock-hands/sequences read as a whole, concave-vs-convex mirror and convex-vs-concave lens swaps under disguised scenarios, virtual images chased onto screens, prism dispersion order with violet bending most, a single colour 'split' or colours 'created', and converge/diverge sorting across all four curved devices. Do not write on this paper — mark answers on your answer sheet.

1. A boy holds a small opaque disc midway between a single tiny lamp and a wall. He then slides only the disc steadily toward the wall (the lamp and wall stay put). As the disc nears the wall, the dark shadow it casts on the wall steadily:

- (A) grows without limit, because the disc is now closer to the wall
- (B) stays exactly the same size, since shadow size never depends on where the disc sits
- (C) shrinks toward the disc's true size, because the edge-grazing rays diverge less over the shorter remaining gap to the wall
- (D) splits into a coloured fringe as the disc approaches the wall

2. A pinhole camera aimed at a tall lit signboard shows the sign inverted on the screen. The owner then doubles the box length (hole-to-screen distance) while keeping the box the same distance from the sign. The picture on the screen becomes:

- (A) smaller and upright, because a longer box turns the image the right way up
- (B) the same size but now upright, since doubling the length cancels the inversion

- (C) larger and still inverted, because lengthening the straight-line ray paths beyond the hole spreads the crossed rays farther apart
- (D) split into colours, because the longer path lets the light disperse

3. Two identical small lamps, side by side, shine past a single thin upright pole onto a wall, producing a shadow with a fully dark central strip flanked by two paler grey strips. The two grey strips exist because, at those wall points:

- (A) only one of the two lamps is hidden by the pole while the other still lights the spot, so the region is partly illuminated
- (B) the pole bends a little light sideways to soften each edge
- (C) the wall itself turns grey wherever a shadow boundary lands
- (D) the light slows down beside the pole and spreads to make a grey fringe

4. Through three opaque cards each pierced with a pinhole, a student can see a distant lamp only when all three holes line up with the eye and lamp. If the middle card alone is shifted a little to one side, the lamp vanishes. The cleanest single principle this proves is that light:

- (A) is reflected from one card to the next until it reaches the eye
- (B) travels in straight lines, so blocking the straight path at any one card cuts off the view
- (C) bends through each pinhole to find its way to the eye
- (D) must pass through holes of exactly equal size to be seen

5. A laser beam crosses a smoky room as a dead-straight bright line. A thin flat glass slab with two parallel faces is slid into its path; on the far side the line continues straight but shifted sideways. The fact that the emerging beam is parallel to the original (just displaced) tells us that across the slab:

- (A) the glass curved the beam permanently, like a lens, into a new direction
- (B) the glass split the beam into colours that then merged back into one direction
- (C) the bending at the first face is undone by an equal, opposite bending at the second parallel face, leaving the direction unchanged
- (D) the beam stopped travelling straight inside the glass and wandered randomly

6. At noon a vertical flagpole casts a short shadow; by late afternoon the shadow is long. Within this chapter, the single fact that makes the shadow lengthen as the afternoon wears on is that:

- (A) the pole physically grows taller as the day cools
- (B) the Sun's light bends around the pole more strongly in the afternoon
- (C) the ground reflects the pole's image outward to make a longer dark patch
- (D) light travels straight from the lower Sun, so the pole's top blocks rays along a more slanted line that meets the ground farther away

7. A single beam of light strikes a plane mirror, and the angle between the incident ray and the reflected ray (the full opening between them) is measured as 110 degrees. The angle of incidence (measured from the normal) is therefore:

- (A) 55 degrees, because the incident and reflected rays are equally inclined to the normal so each takes half of the 110 degrees
- (B) 110 degrees, since the angle between the two rays is the angle of incidence
- (C) 20 degrees, found by subtracting 90 degrees from 110 degrees
- (D) 35 degrees, the angle the incident ray makes with the mirror surface

8. A polished tile and a sheet of matte cardboard are placed side by side in the same parallel sunlight. From the tile you catch a sharp bright glare in one direction; from the cardboard you see an even glow from every direction. At each microscopic point on BOTH surfaces, the law of reflection:

- (A) holds only on the tile; the cardboard breaks the law because it is rough
- (B) holds exactly, but the cardboard's tiny tilted facets send the equally-reflected rays off in many directions, destroying any single image
- (C) fails on both, because rough and smooth surfaces obey different reflection rules
- (D) holds only on the cardboard, because rough surfaces reflect more strongly

9. You can read this printed page even though paper makes no light of its own, yet you cannot see your face in the same page. The single property of the paper that explains BOTH facts at once is that it:

- (A) reflects light regularly, which both lights the print and forms a faint face-image
- (B) produces its own faint light strong enough to read by but too weak to mirror a face
- (C) absorbs all light, so you read by the shadows of the letters
- (D) reflects room light diffusely, scattering it from every point to your eyes so the print is visible but no organised face-image survives

10. A still puddle shows a sharp upside-down image of a building; when a breeze ruffles the water, the image breaks into shimmering patches. The ruffling destroys the image because the disturbed surface now produces:

- (A) refraction, bending the building's light down into the water
- (B) absorption, swallowing the light that used to form the image
- (C) dispersion, splitting the building's light into separate colours
- (D) diffuse reflection, since each tilted ripple sends its reflected rays in a different direction so the rays no longer arrive as one organised image

11. You stand 90 cm in front of a plane mirror holding a photo flat against your chest, 25 cm in front of you (so the photo is 65 cm from the mirror). The distance from the photo to the photo's OWN image is:

- (A) 180 cm, twice your 90 cm distance from the mirror
- (B) 65 cm, equal to the photo's distance from the mirror

- (C) 130 cm, twice the photo's 65 cm distance from the mirror
- (D) 115 cm, the sum of 90 cm and 25 cm

12. A man walks toward a large plane mirror at 1.2 metres per second. Considering both his approach and his image's approach to the mirror, the gap between the man and his image closes at:

- (A) 1.2 metres per second, since only the man is actually moving
- (B) 2.4 metres per second, because the image approaches the mirror at the same 1.2 m/s as the man, from the other side
- (C) 0.6 metres per second, half the man's speed
- (D) the gap stays fixed because the image keeps pace and never gets nearer

13. A 1.5 m tall girl stands before a plane mirror, then steps back until she is four times her original distance from it. The height of her image is now:

- (A) one-quarter of 1.5 m, since she is four times farther
- (B) four times 1.5 m, matching her new distance
- (C) 6 m, equal to the new object-plus-image distance
- (D) still 1.5 m, because a plane-mirror image is always exactly the object's size at any distance

14. A boy stands 2 m from a plane mirror and his sister stands 6 m from the SAME mirror, both directly facing it. The distance between the boy's image and the sister's image, both measured behind the mirror, is:

- (A) 4 m, the difference between their two image distances of 2 m and 6 m behind the mirror
- (B) 8 m, the sum of their two distances from the mirror
- (C) 16 m, the sum of the doubled distances
- (D) 12 m, twice the sister's distance

15. An object 8 cm in front of a plane mirror is moved to 13 cm in front. Compared with before, the image now lies:

- (A) 10 cm farther behind, twice the 5 cm the object moved
- (B) 5 cm closer to the mirror, because moving back brings the image forward
- (C) in the same place, since plane-mirror images do not move
- (D) 5 cm farther behind the mirror, because the image always sits as far behind as the object is in front

16. Two plane mirrors face each other on opposite walls 4 m apart, and a lamp hangs exactly midway between them. Ignoring brightness loss, the nearest image of the lamp formed in EITHER mirror lies:

- (A) 4 m behind that mirror, equal to the full gap between the mirrors
- (B) 1 m behind that mirror, half of the lamp's distance to it

(C) 2 m behind that mirror, because the lamp is 2 m in front of each mirror and the image sits an equal distance behind

(D) at the surface of the mirror, since the lamp is centred

17. The capital word 'PEN' is written on clear film and held facing you up to a plane mirror. The mirror image reads:

(A) exactly 'PEN', unchanged, because the letters are symmetric

(B) 'PEN' upside down, because the mirror flips top and bottom

(C) 'NEP' with each letter still drawn the normal way round

(D) the letters reversed left-to-right and in reverse order, so it looks like a backwards N, then a backwards E, then a backwards P

18. A digital clock built from straight segments shows 21:05. Viewed in a plane mirror that swaps left and right (and assuming each digit's mirror still looks like a digit), the reflected display would appear to read:

(A) 21:05, unchanged, because digits made of straight segments never flip

(B) 20:15, because the whole row reverses order to 50:12 read right-to-left and each segment-digit mirrors to give 20:15

(C) 50:12, simply the characters in reverse with no other change

(D) 12:50, the digits reversed and the colon moved

19. A round wall clock with no numbers shows its hour hand pointing toward the 4-o'clock position (lower right) and its minute hand straight up. In a plane mirror the reflected clock's hour hand points toward the:

(A) 4-o'clock position still, because the hand is below the centre

(B) 10-o'clock position, by flipping top to bottom

(C) 2-o'clock position, by reflecting across the horizontal line

(D) 8-o'clock position (lower left), because lateral inversion mirrors the 4 across the vertical 12-to-6 line to the 8 side

20. A teacher writes a secret note in ordinary handwriting, then holds it up to a plane mirror; a pupil copies down exactly what the mirror shows, letter by letter. The pupil's hand-copied page, held up to a SECOND plane mirror, will read:

(A) doubly reversed and even harder to read than the first reflection

(B) upside down but otherwise the same as the original

(C) the teacher's original note, correctly, because a second lateral inversion undoes the first

(D) still reversed, because two mirrors reverse twice as much

21. Of the capital letters E, H, K, X, exactly which set would look UNCHANGED in a plane mirror, given the mirror swaps left and right (not top and bottom)?

(A) E and K, because they have straight horizontal strokes

(B) all four, since capital letters are always symmetric

- (C) none of them, because every letter is reversed by a mirror
- (D) H and X, because each is symmetric about a vertical line down its middle

22. On the front of an ambulance the word that the chapter discusses is painted so a driver ahead reads it correctly in the rear-view mirror. To achieve this for the word 'AID', it must actually be painted on the bonnet as:

- (A) exactly 'AID' written normally, because the mirror reads words as they are
- (B) each letter reversed left-to-right and the letters in reverse order, so the mirror's lateral inversion turns it back into 'AID'
- (C) 'AID' written upside down so the mirror flips it upright
- (D) 'DIA' with each letter drawn the normal way round

23. A glass door has a message written so people OUTSIDE read it correctly. A student inside therefore sees it reversed through the glass. If that student holds a small plane mirror up to view the reversed message, the mirror will show it:

- (A) doubly reversed, so it becomes even harder to read
- (B) upside down, because the mirror flips the message vertically
- (C) unchanged and still reversed, because a mirror cannot fix reversed text
- (D) correct and readable, because the mirror's lateral inversion cancels the reversal the student was already seeing

24. A magician marks the empty spot behind a plane mirror where a candle's flame appears to hover, then slides a white card to that exact spot, but no picture lands on it. The deepest reason is that:

- (A) no real light ever reaches that spot; the rays only appear to spread from there, so the image is virtual and cannot fall on a card
- (B) the image is real but is formed slightly off the marked spot
- (C) the card is too thin and pale to hold the bright picture
- (D) the image is inverted there and cancels itself out on the card

25. Comparing a plane mirror, a concave mirror used on a very DISTANT object, and a convex mirror, the property that the plane and convex mirrors share but the concave-on-a-distant-object does NOT is:

- (A) forming a virtual image you cannot catch on a screen, whereas the concave mirror's distant-object image is real
- (B) magnifying the object, which the plane and convex do but the concave does not
- (C) forming a real image, which only the plane and convex can do
- (D) inverting the image, which the plane and convex do and the concave does not

26. A polished steel ladle shows your face upside-down and large when you look into its bowl from a little way off, but tiny and upright when you look at its rounded back. The bowl side and the back side are acting, in order, as:

- (A) a convex mirror giving an erect image, and a concave mirror giving an inverted image
- (B) a concave mirror giving a real inverted image, and a convex mirror giving a virtual erect diminished image
- (C) a convex lens and a concave lens
- (D) a plane mirror on the bowl and a concave mirror on the back

27. A student claims, 'A concave mirror always magnifies, and it always makes an upright image.' The correct verdict is:

- (A) both claims are true for every object position
- (B) the magnifying claim is true but the upright claim is false
- (C) the upright claim is true but the magnifying claim is false
- (D) both claims are false; a concave mirror gives a small inverted real image of a distant object and an enlarged erect virtual image only when the object is very close

28. A solar cooker focuses sunlight to heat a pot, a car headlight sends out a strong forward beam, and a dentist's mirror shows an enlarged upright tooth. The single mirror TYPE common to all three is:

- (A) the concave mirror, which converges sunrays to a hot point, throws a near-parallel beam when the bulb is at its focus, and enlarges a close tooth uprightly
- (B) the convex mirror, used in all three for its wide view
- (C) the plane mirror, used in all three to keep true sizes
- (D) a mix: the cooker uses convex, the headlight plane, the dentist concave

29. As an object is carried steadily from very far away right up to the face of a concave mirror, the image it forms changes in the order:

- (A) virtual and erect while far, then real and inverted once close
- (B) real and inverted throughout, at every distance
- (C) real and inverted while the object is far, then becoming virtual, erect and enlarged once the object is very close
- (D) virtual and diminished throughout, never changing

30. A torch maker wants the brightest, narrowest forward beam. He must place the bulb at the focus of the reflector. The correct reflector, and what it does there, is:

- (A) a convex mirror, which spreads the rays into a wide beam for safety
- (B) a plane mirror, which keeps the beam exactly the same width as the bulb
- (C) a concave mirror, which reflects the rays from a focus-placed bulb as a near-parallel forward beam
- (D) a convex lens used as a mirror, which converges the beam to a single point ahead

31. Held a few millimetres from a back tooth, a dentist's mirror shows the tooth enlarged and the right way up; aimed instead at the far wall it would show a small upside-down image. These two behaviours together identify the mirror as:

- (A) convex, because it magnifies the close tooth and diminishes the far wall
- (B) plane, because it shows both the tooth and the wall the right way up
- (C) concave, giving a virtual erect magnified image of the very close tooth and a real inverted diminished image of the distant wall
- (D) a convex lens, because it can both magnify and invert depending on distance

32. A shop owner mounts a curved mirror high in a corner and is delighted that 'one small camera now watches the whole floor, though everyone looks small.' The mirror that gives a WIDE field of view with everyone erect and diminished is:

- (A) the concave mirror, because it gathers a wide view to a focus
- (B) the plane mirror, because flat mirrors give the widest possible view
- (C) the concave mirror, because it magnifies distant shoppers so they stand out
- (D) the convex mirror, whose diverging reflection always forms a small, erect, virtual image while taking in a wide area

33. A driver wants ONE side mirror that shows the widest stretch of road behind AND keeps every following vehicle right-side-up. The correct mirror, and the reason a concave one would fail, is:

- (A) a convex mirror, because it gives a wide erect view, whereas a concave mirror could invert distant vehicles and narrows the view
- (B) a concave mirror, because it gives both a wide view and an erect image
- (C) a plane mirror, because it gives the widest view of all and keeps vehicles erect
- (D) a concave lens fitted as a mirror, because it diverges light for a wide view

34. A jeweller picks up a hand-held disc of glass that is thicker in the middle than at the rim, holds it close above a gem, and sees the gem enlarged and upright; the SAME disc held far from a window throws a small upside-down picture of the trees on a wall. The disc is a:

- (A) concave lens, which can give both kinds of image depending on distance
- (B) convex lens, which gives an enlarged erect virtual image close up and a small inverted real image of distant objects
- (C) concave mirror, which reflects to give both images
- (D) convex mirror, which magnifies close and inverts far objects

35. A convex lens held far from a bright window projects a tiny upside-down picture of the trees onto a wall. The same lens then held close above tiny print shows the print enlarged and upright. The two images are, in order:

- (A) both real, since a convex lens always makes real images
- (B) virtual for the trees and real for the print
- (C) real and inverted for the distant trees, then virtual and erect for the close print
- (D) both virtual, since a lens never makes a screen image

36. Sunlight focused by a convex lens can scorch a dot in a leaf, but a same-sized concave lens held in the same sunbeam never scorches anything. The concave lens fails specifically because it:

- (A) diverges the parallel sunrays, spreading them apart so they never gather to a hot point
- (B) reflects the sunlight back toward the Sun instead of letting it through
- (C) absorbs all the sunlight before any can pass through the glass
- (D) splits the sunlight into a harmless cool rainbow band

37. Three transparent discs are offered: P focuses the Sun to a burning spot, Q makes close print look larger and upright, R makes a distant scene appear small and upright. The discs that are CONVEX (converging) lenses are:

- (A) all three, since any lens that changes an image is converging
- (B) P and R, because both shrink what they show
- (C) Q and R, because both give upright images
- (D) P and Q, because both converge light, while R's small erect image marks it as a concave lens

38. Looking at a printed page through a disc of glass thinner in the middle than at its rim, the letters appear smaller and upright no matter where you hold it. The disc is a:

- (A) convex lens, which gives an enlarged erect image
- (B) concave mirror, which reflects an erect image
- (C) concave lens, which always gives a diminished, erect, virtual image
- (D) convex mirror, whose diminished image proves it is a mirror

39. Four devices are sorted by whether they bring parallel rays TOGETHER or spread them APART: a concave mirror, a convex mirror, a convex lens, a concave lens. The two that CONVERGE light are:

- (A) the concave mirror and the convex lens
- (B) the convex mirror and the concave lens
- (C) the concave mirror and the concave lens
- (D) the convex mirror and the convex lens

40. A teacher pairs each device with its shape. Which row is entirely correct?

- (A) convex lens thinner in the middle; concave lens thicker in the middle; concave mirror bulging outward; convex mirror curved inward
- (B) convex lens thicker in the middle; concave lens thinner in the middle; concave mirror curved inward; convex mirror bulging outward
- (C) convex lens curved inward; concave mirror thicker in the middle; convex mirror thinner in the middle; concave lens flat
- (D) all four are simply flat plates of glass or metal

41. A student is handed a real, erect, enlarged image and asked which single mirror from this chapter could have produced it. The honest answer is:

- (A) no mirror in this chapter can, because any real mirror image is inverted, never erect
- (B) a concave mirror with the object held very close to it
- (C) a convex mirror, which enlarges nearby objects
- (D) a plane mirror, which keeps the image erect and can make it larger

42. Which row correctly states the kind of image each device forms?

- (A) concave mirror on a distant object gives real inverted; convex mirror gives virtual erect diminished; convex lens on a distant object gives real inverted
- (B) plane mirror gives real same-size; convex mirror gives virtual magnified; concave lens gives real
- (C) concave mirror always gives virtual erect; convex lens always gives virtual; concave lens gives real
- (D) convex mirror gives real on a screen; plane mirror gives inverted; concave mirror never gives a real image

43. A complete statement about converging versus diverging behaviour is required. Which one is correct throughout?

- (A) a concave mirror and a concave lens both converge, while the convex ones both diverge
- (B) a concave mirror and a convex lens both converge light, while a convex mirror and a concave lens both diverge it
- (C) a convex mirror and a convex lens both converge, while the concave ones both diverge
- (D) all four devices converge light to a single point

44. A quiz card lists three device-to-image pairings and asks which CARD has all three right: which one is fully correct?

- (A) plane mirror to real inverted; convex mirror to real magnified; concave lens to real
- (B) plane mirror to virtual same-size; concave lens to virtual diminished; convex lens on a distant object to real inverted
- (C) concave mirror always to virtual erect; convex lens always to virtual; concave lens to real
- (D) convex mirror to a real screen image; plane mirror to an inverted image; concave mirror never to a real image

45. A child must name the device that, like a plane mirror, always gives an ERECT image, yet unlike the plane mirror always gives a SMALLER image whatever the object's position. The device is a:

- (A) concave mirror used on a near object, which is erect and magnified
- (B) concave mirror used on a distant object, which is erect and diminished
- (C) convex mirror, which is always erect like a plane mirror but always diminished
- (D) concave lens, which is a lens rather than a mirror

- 46.** Which single image can actually be caught on a sheet of paper held in the right place?
- (A) your face seen in a flat bathroom mirror
 - (B) the wide view of a street seen in a car's convex side mirror
 - (C) the enlarged print seen through a magnifying glass held close to it
 - (D) the small upside-down picture of a faraway window formed by a convex lens
- 47.** A teacher demonstrates: (i) a calm lake reflecting a hill, (ii) sunlight through a prism making a colour band, (iii) a hand lens throwing an upside-down picture of a window on a wall. These illustrate, in order:
- (A) dispersion, reflection, and a virtual image from a concave lens
 - (B) reflection, dispersion, and a real image formed by a convex lens
 - (C) refraction, dispersion, and reflection
 - (D) diffuse reflection, lateral inversion, and magnification
- 48.** A camera lens that sharply focuses a far scene onto film, a magnifying glass, and a torch's curved reflector are grouped by what they do to parallel rays. Which statement is correct?
- (A) all three converge parallel rays, since the camera lens and magnifying glass are convex lenses and the torch reflector is a concave mirror
 - (B) the magnifying glass is the odd one out because it only diverges light
 - (C) the camera lens is the odd one out because it spreads light out
 - (D) the torch reflector is the odd one out because it scatters light randomly
- 49.** Two clear glass blocks sit in the same sunbeam: one is a triangular prism, the other a flat-faced cube. Only the prism casts a band of colours. The single geometric difference that lets the prism disperse while the cube does not is that the prism's two light-bending faces:
- (A) are tinted, so they add the colours to the otherwise plain white light
 - (B) are made from a rare rainbow glass that the cube lacks
 - (C) are gently curved like a lens, and the curve is what creates the colours
 - (D) meet at an angle (are not parallel), so the colours separated at the first face keep diverging through the second rather than recombining
- 50.** Reading a prism's spectrum from the colour bent LEAST to the colour bent MOST, the correct order is:
- (A) red, orange, yellow, green, blue, indigo, violet
 - (B) violet, indigo, blue, green, yellow, orange, red
 - (C) red, yellow, blue, then back to violet only
 - (D) green, red, violet, blue, orange, yellow, indigo
- 51.** Pure indigo light, a single colour, is passed through a glass prism. What emerges is:
- (A) the full seven-colour spectrum, because every prism makes all seven colours
 - (B) white light, because the prism recombines the colours

- (C) no light at all, because a prism blocks single colours
- (D) only indigo light, bent through the prism but not split into any other colours

52. A child catches a prism's colour band on a card, then passes that band through a SECOND, reversed prism placed just behind the first; the colours merge and a white patch reappears on the wall. What does this two-prism result establish?

- (A) the first prism manufactured the colours and the second one destroyed them
- (B) the colours were already present in the white light all along, since separating then recombining them gives white back
- (C) white light is a single pure colour that the prisms only tinted
- (D) the second prism added new colours that happened to look white

53. After a rain shower, with the low Sun behind you, a rainbow arches across the sky. A teacher says, 'The raindrops paint new colours onto the sky.' The correct response is:

- (A) yes, the raindrops manufacture seven fresh colours from clear sunlight
- (B) yes, but the raindrops make only red and violet, and the rest come from the sky
- (C) no new colours are made; each raindrop acts like a tiny prism, separating colours already present in the white sunlight
- (D) the rainbow's colours come from the blue of the sky, not from the Sun

54. Two seven-colour discs are spun side by side, one fast and one barely turning. The fast one looks whitish while the slow one still shows streaks of colour. The reason the SLOWER disc fails to look white is that:

- (A) slow spinning makes the painted colours fade away
- (B) its colours arrive at the eye slowly enough to be told apart, so the eye does not blend them into white
- (C) only fast motion turns the seven paints physically into white paint
- (D) the slow disc reflects extra colours that the fast one does not

55. Three demonstrations are shown: (i) a hand lens projecting an upside-down picture of a distant window on a wall, (ii) sunlight scattered evenly off a sheet of chalk-white card, (iii) a digital clock read backwards in a mirror. They illustrate, in order:

- (A) dispersion, regular reflection, and refraction
- (B) a virtual image from a concave lens, absorption, and dispersion
- (C) a real image from a convex lens, diffuse reflection, and lateral inversion
- (D) diffuse reflection, a real image from a convex lens, and dispersion

56. A student must pick the ONE description that is true of a convex mirror for every object position and correctly explains its everyday use. The right pick is:

- (A) it converges reflected rays into a real inverted image, which is why it lines a torch reflector
- (B) it diverges reflected rays, giving an always-virtual, erect, diminished image and a wide field of view, which is why it serves as a rear-view mirror

- (C) it enlarges nearby objects, which is why a dentist holds it close to a tooth
- (D) it forms a real image on a screen when the object is far away

57. A make-up artist wants a mirror giving an enlarged upright face when held close, a delivery van wants one showing a wide stretch of road behind, and a solar engineer wants one focusing sunlight to a hot point. The three mirrors needed, in that order, are:

- (A) convex for the face, concave for the rear view, and convex for the solar focus
- (B) concave for the face, plane for the rear view, and convex for the solar focus
- (C) concave for the close enlarged face, convex for the wide rear view, and concave for the solar focus
- (D) plane for the face, convex for the rear view, and plane for the solar focus

58. A device gives an enlarged upright image of close print AND can throw a small upside-down picture of a distant lamp onto a wall, BUT it is a lens, not a mirror. The device, and why it is not a concave mirror, is:

- (A) a convex lens, because light passes THROUGH it to form the images, whereas a concave mirror would reflect the light back
- (B) a concave lens, because only a concave lens does both kinds of image
- (C) a concave mirror, because mirrors give both close magnified and far inverted images
- (D) a convex mirror, because it transmits light to form both images

59. A pupil sorts five claims about mirrors and lenses and asks which one is the ONLY FALSE statement. Which is false?

- (A) a convex lens always forms a virtual image whatever the object position
- (B) a plane mirror image is virtual and the same size as the object
- (C) a convex mirror always forms a virtual diminished image
- (D) a concave mirror forms a real inverted image of a distant object

60. A frosted tube light (a broad source) shines past a small marble onto a wall. The shadow has a pitch-dark centre but a soft grey halo around it, unlike the crisp-edged shadow the same marble made under a tiny torch bulb. The grey halo appears under the tube light because, at those halo points:

- (A) the marble bends a little light around its edge to soften the boundary
- (B) the marble hides only part of the broad tube from view, so the unblocked part of the tube still throws some light there
- (C) the wall colours the light grey wherever a shadow edge falls
- (D) the light slows as it passes the marble and smears into a grey fringe

Solutions — Science (Class 7), Chapter 15: Light — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	D	31	C	41	A	51	D
2	C	12	B	22	B	32	D	42	A	52	B
3	A	13	D	23	D	33	A	43	B	53	C
4	B	14	A	24	A	34	B	44	B	54	B
5	C	15	D	25	A	35	C	45	C	55	C
6	D	16	C	26	B	36	A	46	D	56	B
7	A	17	D	27	D	37	D	47	B	57	C
8	B	18	B	28	A	38	C	48	A	58	A
9	D	19	D	29	C	39	A	49	D	59	A
10	D	20	C	30	C	40	B	50	A	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) With a near-point source, rays graze the disc's rim and continue straight, spreading apart on their way to the wall; the farther the wall lies beyond the disc, the more those rays have spread, so the shadow is largest when the disc is near the lamp and smallest (close to the disc's own size) when the disc is pressed near the wall. Moving the disc toward the wall therefore shrinks the shadow. It cannot grow without limit — that would be the case if the disc moved toward the lamp, not the wall. Shadow size plainly does depend on positions, so 'never depends' is false. A shadow is an absence of light, not a refracted band, so no coloured fringe is produced by an opaque disc in straight-line light.

2. (C) Light travels straight, so a ray from the top of the sign passes through the pinhole and lands low on the screen, and a bottom ray lands high — that crossing is what inverts the picture and it happens at the hole regardless of box length. Making the screen farther from the hole lets the already-crossed rays continue diverging, so the inverted picture grows larger. Box length never flips the image upright; the inversion is fixed by the single crossing at the hole. A pinhole does not act like a prism, so no dispersion band appears.

3. (A) Each lamp casts its own full shadow of the pole; where the two full shadows overlap, both lamps are blocked and the strip is fully dark, but in the flanking strips one lamp's rays are blocked while the other lamp's rays still arrive, so those points receive half the light and look grey. Light travels straight and an opaque pole does not bend rays around itself. The wall's colour is fixed; it does not 'turn grey' — it simply receives less light. Light does not slow and smear beside a solid object in this chapter's straight-line model.

4. (B) Because light goes straight, only rays lying on the single straight line through all three holes, the eye and the lamp can reach the eye; sliding the middle card breaks that line and no straight path remains, so the lamp disappears. The light is not bounced card-to-card — these are holes, not mirrors. If light bent through pinholes it could curve around the shifted card and still be seen, but it cannot. Equal hole size is irrelevant; alignment, not size, is what matters.

5. (C) A flat slab has two parallel faces; the ray bends one way entering and the equal-but-opposite amount leaving, so it emerges travelling in its original direction merely offset to the side — hence the line stays parallel. A lens has non-parallel curved faces and would change direction, which the slab does not. Any slight colour separation in a parallel slab recombines because the faces are parallel, but the question is about direction, and the beam stays one straight line. Inside any uniform medium light still travels straight, not randomly.

6. (D) Sunlight comes in straight rays; when the Sun is high the rays graze the pole top almost straight down, making a short shadow, but when the Sun is low the grazing ray slants and travels far across the ground before it would have hit, so the shadow is long. The pole's height is fixed. Light does not bend around the pole — a shadow is precisely where straight rays are blocked. A shadow is missing light, not a reflected image cast by the ground.

7. (A) The normal bisects the angle between the incident and reflected rays because both make equal angles with it; so each ray is at half of 110 degrees, i.e. 55 degrees from the normal, and the angle of incidence is 55 degrees. The full 110 degrees is the sum of the two equal angles, not one of them. Subtracting 90 has no physical basis here. The 35 degrees would be the angle to the mirror surface (90 minus 55), not the angle of incidence, which is measured from the normal.

8. (B) Every individual ray obeys angle-in-equals-angle-out at the point it strikes, on smooth and rough surfaces alike; the cardboard merely presents countless tiny facets pointing different ways, so the obediently-reflected rays scatter and no organised image forms — this is diffuse reflection. The law is universal, so it does not 'fail' on the rough surface. There are not two different reflection rules. Roughness scatters light, it does not reflect more strongly.

9. (D) Paper is rough, so it scatters incoming room light in all directions (diffuse reflection): that scattered light reaches your eyes from every part of the page, letting you read the print, but because the rays are sent every which way no coherent image of your face can form. Regular reflection would give a face-image, which paper does not. Paper emits no light of its own. If it absorbed all light you would see only black, not white print on a page.

10. (D) A calm surface acts like a flat mirror giving regular reflection and a clean image; ripples tilt the surface into many small slopes, so reflected rays scatter in many directions — diffuse reflection — and the single organised image dissolves into shimmer. The light is reflected, not bent down into the water (refraction) nor swallowed (absorption). No prism-like splitting into colours is involved; the image simply scatters.

11. (C) A plane mirror forms each object's image as far behind the mirror as the object is in front, so the photo (65 cm in front) has its image 65 cm behind, making object-to-image $65 + 65 = 130$ cm. Using your own 90 cm gives your image distance, not the photo's. 65 cm is only the one-way distance, not the round trip to the image. Adding 90 and 25 mixes unrelated distances and has no meaning here.

12. (B) The image is always as far behind the mirror as the man is in front, so as he closes in at 1.2 m/s his image closes in at 1.2 m/s from the other side; the man-to-image gap therefore shrinks at $1.2 + 1.2 = 2.4$ m/s. Counting only the man ignores that the image also moves. Halving the speed has no basis. The gap does change — it shrinks twice as fast as the man moves, it does not stay fixed.

13. (D) A plane mirror always makes an image the same size as the object no matter how near or far it stands; distance only changes how far behind the mirror the image sits, never its size, so the image stays 1.5 m tall. It does not shrink with distance — that confuses how big the image looks (its apparent size) with its true size. It cannot grow to match distance. The 6 m figure mixes the round-trip distance with height, which are unrelated.

14. (A) Each image lies as far behind the mirror as its owner stands in front: the boy's image is 2 m behind, the sister's is 6 m behind, both on the same side, so the gap between the two images is $6 - 2 = 4$ m. Adding 2 and 6 would be right only if the images sat on opposite sides of the mirror, which they do not. Doubling the distances is the object-to-image trip, not the image-to-image gap. Twice the sister's distance ignores the boy entirely.

15. (D) The image distance behind equals the object distance in front; moving the object from 8 cm to 13 cm shifts the image from 8 cm behind to 13 cm behind, i.e. 5 cm farther back — the same 5 cm the object moved. The image moves the same amount as the object, not double. Moving the object away moves the image away too, not closer. The image certainly moves when the object moves; it tracks one-to-one.

16. (C) The lamp sits midway, so it is $4 \div 2 = 2$ m in front of each mirror; a plane mirror puts the image as far behind as the object is in front, so the first image is 2 m behind. The full 4 m is the mirror-to-mirror gap, not the image distance. Halving again to 1 m has no basis. The image is not at the surface — it is 2 m behind, matching the object's 2 m in front.

17. (D) A plane mirror laterally inverts (swaps left and right) and reverses the reading order, so 'PEN' becomes N-E-P with each letter also flipped about a vertical line: P, E and N each bulge or open to one side, so none is symmetric and all appear backwards. It is not unchanged — every letter here is asymmetric. The mirror swaps left-right, not top-bottom, so it is not upside down. The reading order does reverse to N-E-P, but each letter is flipped too, so 'NEP' drawn the normal way round is wrong.

18. (B) Lateral inversion reverses the left-to-right order of the whole display and mirrors each symbol. Reversing 2 1 : 0 5 gives the visual sequence 5 0 : 1 2, and mirroring those seven-segment shapes turns 5 into 2 and 2 into 5 (1 and 0 stay), yielding 2 0 : 1 5, i.e. 20:15. It is not unchanged — 2 and 5 are not vertically symmetric. Plain '50:12' forgets that 5 and 2 themselves flip in the mirror. '12:50' wrongly keeps the colon's neighbours and mis-mirrors the digits.

19. (D) A plane mirror flips left and right about the vertical axis (the 12-6 line), so a hand at 4 (lower right) reflects to the lower left, which on a clock face is the 8 position. It does not stay at 4 — only hands lying on the 12 or 6 line are unchanged. The mirror does not flip top-to-bottom, so 10 is wrong. Reflecting across the horizontal axis would give 8 only by coincidence of reasoning; the correct mirror axis here is vertical, and 4 maps to 8, while a horizontal flip would send 4 to 2, which is the wrong axis.

20. (C) Each plane mirror laterally inverts once; the pupil copied a once-reversed version, and viewing that in a second mirror reverses it again, and two reversals cancel, restoring the original readable note. Two reversals do not add up to 'more reversed' — they undo each other. Mirrors swap left-right, not top-bottom, so it is not upside down. 'Still reversed' would be true after one mirror, but the second mirror flips it back.

21. (D) A plane mirror reverses left and right, so a letter is unchanged only if its left half mirrors its right half about a vertical centre line: H and X satisfy this, but E and K bulge only to the right and so flip into backwards shapes. E and K are NOT vertically symmetric, so that set is wrong. Not all capitals are symmetric — most are not. Vertically symmetric letters like H and X do survive unchanged, so 'none' is false.

22. (B) A rear-view mirror laterally inverts, so to be read as 'AID' it must be painted pre-reversed: the letters in reverse order AND each letter mirror-flipped, so that the mirror's flip undoes it to 'AID'. Writing it normally would appear reversed in the mirror. Upside-down writing is wrong because the mirror swaps left-right, not top-bottom. 'DIA' with normally-drawn letters reverses only the order, not each letter's handedness, so the mirror would not restore 'AID'.

23. (D) The student sees the writing reversed (one flip). A plane mirror applies a second left-right flip, and two flips cancel, so the mirror shows the message the right way round and readable. It is not doubly reversed — the two reversals undo each other, not stack. A mirror swaps left-right, not top-bottom, so it is not upside down. A mirror absolutely can fix already-reversed text, which is exactly what happens here.

24. (A) A plane mirror makes a virtual image: the reflected rays in front of the mirror only seem to diverge from a point behind it, but no light actually crosses that point, so a card placed there catches nothing. The image is not merely mislocated — it is virtual everywhere behind the mirror. Card thickness or paleness is irrelevant; even a perfect screen catches nothing. A plane-mirror image is erect, not inverted, and 'cancelling out' is not why screens fail — the absence of real light is.

25. (A) A plane mirror and a convex mirror always give virtual, screen-uncatchable images, while a concave mirror viewing a distant object gives a real, inverted, screen-catchable image — so the shared 'virtual' property excludes the concave-distant case. The plane mirror does not magnify (same size) and the convex diminishes, so 'magnify' is wrong. It is the concave, not the plane/convex, that makes the real image, so 'only plane and convex form real' is backwards. The plane mirror gives an erect, not inverted, image, so 'plane and convex invert' is false.

26. (B) The inward-curving bowl is a concave mirror; viewed from beyond its focus it forms a real, inverted, image (here enlarged) — matching the upside-down face. The outward-bulging back is a convex mirror, which always gives a virtual, erect, diminished image — matching the tiny upright face. The two are not swapped; bowl is concave (inverted), back is convex (erect). A ladle reflects, it is not a lens. The bowl is not flat, so it is not a plane mirror.

27. (D) A concave mirror's image depends on where the object sits: a distant object gives a diminished, inverted, real image, while a very close object gives an enlarged, erect, virtual image — so neither 'always magnifies' nor 'always upright' holds. Both are therefore false. Claiming 'always magnifies' fails for the distant small image; claiming 'always upright' fails for the inverted distant image; so neither half of the original statement survives.

28. (A) All three rely on the converging power of a concave mirror: it brings parallel sunrays to a hot focus (cooker), reverses that to a parallel beam when the source sits at the focus (headlight), and held close gives an enlarged erect virtual image (dentist). A convex mirror only diverges and diminishes, useless for focusing or beaming. A plane mirror neither focuses nor enlarges. The three are not a mix — they are all concave.

29. (C) A concave mirror gives a real, inverted image of a distant object; as the object approaches and finally comes very close to the mirror, the image switches to virtual, erect and enlarged — the shaving-mirror effect. The order is real-then-virtual, not the reverse. It

does not stay real for all distances (close up it is virtual). It is not always diminished — that describes a convex mirror, not a concave one.

30. (C) A concave mirror with the bulb at its focus reverses the focusing action: rays leaving the focus reflect off the curved surface as a parallel (or near-parallel) bundle, giving a tight forward beam. A convex mirror diverges light into a wide weak spread, the opposite of what a torch needs. A plane mirror cannot collimate a diverging bulb into a beam. A lens is not a mirror, and converging to a single point ahead would not give a long straight beam down the path.

31. (C) Only a concave mirror does BOTH: enlarged erect virtual image when the object is very close (the tooth) and small inverted real image when the object is distant (the wall). A convex mirror always diminishes and never inverts, so it cannot give the small inverted far image. A plane mirror never magnifies nor inverts. A lens transmits light, not a mirror that reflects it; the device here reflects off a curved surface.

32. (D) A convex mirror bulges outward and diverges reflected rays, so it always forms a virtual, erect, diminished image and sweeps in a wide field of view — ideal for corner security mirrors where everyone looks small. A concave mirror converges and can invert/enlarge, narrowing rather than widening the useful view. A flat mirror's field of view is no wider than the same-size convex mirror's; in fact it is narrower. A concave mirror would magnify, not give the wide diminished view described.

33. (A) A convex mirror always gives an erect, diminished, virtual image and a wide field of view, so distant cars stay upright and a broad stretch is visible — exactly what is needed. A concave mirror forms a real inverted image of distant objects (cars upside down) and a narrower useful field, so it fails the 'erect' and 'wide' requirements. A plane mirror's view is narrower than the convex's for the same size. A concave lens is not a mirror and cannot serve as the reflecting side mirror.

34. (B) Thicker-in-the-middle glass is a convex (converging) lens: held close it magnifies an erect virtual image (a magnifying glass), and far from a bright distant scene it focuses a small inverted real image onto a wall — both behaviours are signatures of a convex lens. A concave lens is thinner in the middle and only ever gives a small erect virtual image, never an inverted real one. The disc is transparent glass that light passes through, so it is a lens, not a mirror; a convex mirror reflects and never throws a real image on a wall.

35. (C) A convex lens forms a real, inverted image of a distant object (the wall picture you can catch on a screen) and a virtual, erect, enlarged image of a very close object (the magnifying-glass effect). So the order is real-inverted then virtual-erect. It is not 'both real' — the close magnified print is virtual. It is not reversed (trees virtual, print real). And a convex lens definitely can make a screen image of distant objects, so 'both virtual' is wrong.

36. (A) A concave (diverging) lens is thinner in the middle and spreads parallel rays outward, so they leave more spread out and never converge to a concentrated hot spot — hence no scorching. It transmits light, it does not reflect it back like a mirror. It does not absorb all the light — you can see through it. A simple lens does not disperse sunlight into a rainbow; that is a prism's job, and even so dispersion would not stop heating by itself — the real reason is the spreading of the rays.

37. (D) A convex lens converges light, so it can both burn the Sun to a spot (P) and act as a magnifier giving an enlarged erect image close up (Q). R, which makes a distant scene look small and upright, behaves like a concave (diverging) lens, whose image is always diminished and erect — so R is not convex. 'All three' wrongly lumps the diverging R in. 'P and R' pairs a converging lens with a diverging one. 'Q and R' wrongly groups by 'upright' — both convex (close) and concave lenses can look upright, so uprightness alone does not identify converging lenses; the burning spot and magnification do.

38. (C) Thinner-in-the-middle glass is a concave (diverging) lens, and it always forms a diminished, erect, virtual image whatever the object's position — matching the small upright letters. A convex lens (thicker in the middle) magnifies close print, not shrinks it. A concave mirror reflects and would invert a distant object, and is not transparent glass you look THROUGH. A convex mirror also reflects rather than transmits; here you are looking through the glass, so it is a lens, not a mirror.

39. (A) Converging devices bring parallel rays to a focus: a concave mirror (curved inward) and a convex lens (thicker in the middle) both do this. The convex mirror and the concave lens both diverge, spreading rays apart, so they are the non-converging pair. 'Concave mirror and concave lens' mixes one converger with one diverger. 'Convex mirror and convex lens' likewise pairs a diverger with a converger. The clean rule: concave MIRROR and convex LENS converge; convex MIRROR and concave LENS diverge.

40. (B) Correct shapes: a convex lens is thicker in the middle, a concave lens is thinner in the middle, a concave mirror curves inward (like a bowl), and a convex mirror bulges outward (like a dome). The first distractor swaps every lens and mirror descriptor. The second nonsensically assigns lens thicknesses to mirrors and 'flat' to a concave lens. None of these four curved devices is a flat plate, so 'all flat' is false.

41. (A) A real image from any mirror in this chapter is always inverted, so a real, erect, enlarged image simply cannot be made by one mirror — the description is self-contradictory. A concave mirror held very close gives an enlarged erect image, but that one is VIRTUAL, not real. A convex mirror always diminishes, never enlarges. A plane mirror gives a same-size virtual image and cannot enlarge or make a real image. Hence none qualifies.

42. (A) All three parts of the correct row hold: a concave mirror gives a real inverted image of a distant object, a convex mirror always gives a virtual erect diminished image,

and a convex lens gives a real inverted image of a distant object. The first wrong row makes a plane mirror's image 'real' (it is virtual) and a convex mirror 'magnified' (it diminishes). The second wrongly says a concave mirror 'always' virtual (it gives real for distant objects) and a concave lens 'real' (always virtual). The fourth claims a convex mirror gives a screen image (never) and that a concave mirror is never real (it is, for distant objects).

43. (B) The clean pairing is: concave MIRROR + convex LENS converge; convex MIRROR + concave LENS diverge. Notice the mirror and lens that converge have OPPOSITE names (concave/convex) — a classic trap. 'Concave mirror and concave lens converge' is wrong because a concave lens diverges. 'Convex mirror and convex lens converge' is wrong because a convex mirror diverges. Not all four converge — two of them spread light apart.

44. (B) Each pairing on the correct card holds: a plane mirror gives a virtual same-size image, a concave lens gives a virtual diminished image at any distance, and a convex lens gives a real inverted image of a distant object. The first wrong card falsely calls a plane mirror 'real inverted' and a convex mirror 'real magnified.' The second falsely says a concave mirror is 'always' virtual and a concave lens 'real.' The third falsely claims a convex mirror makes a real screen image and that a concave mirror is never real.

45. (C) A convex mirror always forms an erect, virtual, diminished image regardless of object position — erect like a plane mirror but smaller, which is exactly the description. A concave mirror used near gives an enlarged erect image (not smaller). A concave mirror used on a distant object gives an INVERTED real image, not erect. A concave lens is diminished and erect but it is a lens, not a mirror, and the question asks for a mirror.

46. (D) Only a real image can land on a screen, and a convex lens forms a real inverted image of a distant window — catchable on paper. A flat bathroom mirror gives a virtual image (behind the mirror, no real light there). A convex side mirror always gives a virtual image. A magnifying glass held close gives a virtual enlarged image. So only the convex-lens picture of the distant window is real and catchable.

47. (B) A calm lake mirroring a hill is regular reflection; sunlight split by a prism is dispersion; a lens projecting an inverted picture of a distant window on a wall is a real image formed by a convex lens. The order is reflection, dispersion, real-image-from-convex-lens. The lake is not dispersion; the prism is not reflection; and the wall picture is a real image from a convex lens, not a virtual one from a concave lens. The third option mislabels all three phenomena.

48. (A) A camera lens and a magnifying glass are both convex (converging) lenses, and a torch reflector is a concave (converging) mirror — all three bring parallel rays toward a focus, so none is odd; they all converge. The magnifying glass does not diverge — it

converges. The camera lens focuses, it does not spread light out. The torch reflector converges to a beam, it does not scatter randomly.

49. (D) The prism's two faces are inclined to each other; each colour deviates by a slightly different amount, and because the faces are NOT parallel the colours that begin separating at the first face keep diverging through the second, so a spread band emerges. The cube's parallel faces let any separation recombine into white. The faces are clear glass, not tinted — colour is separated, not added. Ordinary glass disperses, so no rare rainbow glass is needed. Prism faces are flat, not curved like a lens, and curvature is not what produces the colours.

50. (A) In a prism red bends the least and violet bends the most, so from least- to most-bent the order is red, orange, yellow, green, blue, indigo, violet (the forward VIBGYOR list read from the red end). The reverse list (violet first) would be most-bent to least-bent, not least to most. 'Red, yellow, blue' omits four colours. The scrambled green-red-violet sequence is not the spectral order at all.

51. (D) A prism disperses white light by separating the many colours already mixed in it; a single pure colour like indigo has nothing to separate, so it simply bends and emerges as indigo. A prism cannot manufacture the other six colours from one. It cannot turn one colour into white. And it does not block colours — the indigo passes through, merely deviated.

52. (B) The first prism spreads white light into its colours and the second, reversed, prism brings them back together into white — recombining the very colours that were separated. This proves the colours were components of the white light to begin with, not anything the prisms made or destroyed. The first prism only separates, it does not manufacture colours; the second only recombines, it does not destroy them. White cannot be a single pure colour if a prism splits it. The second prism adds nothing — it merges the existing colours back to white.

53. (C) Each raindrop disperses sunlight like a small prism, separating the seven colours that white sunlight already contains — no colour is created. Raindrops cannot manufacture colours; they only spread existing ones. They do not make 'only red and violet.' The colours come from dispersed white sunlight, not from the blue sky. So the rainbow reveals, rather than creates, the colours in sunlight.

54. (B) White appears only when the colours reach the eye fast enough that the eye cannot separate them and blends them together; a slowly turning disc lets each colour register on its own, so streaks remain visible. The paints do not fade with slow spinning — they are simply seen separately. Spinning does not convert paint into white paint; the blending happens in the eye, not on the disc. The slow disc reflects the same painted colours as the fast one — it does not gain extra colours.

55. (C) Projecting an inverted picture of a distant window onto a wall is a real image from a convex lens; even scattering off chalk-white card is diffuse reflection; a clock read backwards in a mirror is lateral inversion. The first wrong option mislabels all three. The second calls the lens image virtual-from-concave (it is real-from-convex), the card 'absorption' (it scatters, not absorbs), and the clock 'dispersion'. The third shuffles the labels so none lines up with its demonstration.

56. (B) A convex mirror bulges outward and diverges reflected rays, so for any object it gives a virtual, erect, diminished image with a wide field of view — exactly why vehicles use it for the rear view. It does not converge rays; converging belongs to the concave mirror used in torch reflectors. It does not enlarge; a dentist uses a CONCAVE mirror for close magnification. It never forms a real screen image — its images are always virtual.

57. (C) A close enlarged upright face needs a concave mirror (its near-object virtual magnified image); a wide rear view needs a convex mirror (wide field, diminished erect); focusing sunlight to a hot point needs a concave mirror (converging). So the order is concave, convex, concave. A convex mirror cannot enlarge the face, and a concave mirror would not give the wide erect rear view. A plane mirror neither magnifies the face nor focuses sunlight, and a convex mirror diverges sunlight rather than focusing it.

58. (A) Both behaviours — magnifying close print and projecting a far lamp inverted on a wall — are signatures of a convex lens, and because the light passes through the glass to form the images on the far side, it is a lens, not a reflecting mirror. A concave lens cannot project a real inverted image; it only gives small erect virtual ones. A concave mirror would indeed give both image types, but it reflects rather than transmits, so it is excluded by 'it is a lens, not a mirror.' A convex mirror neither magnifies nor projects a real image, and a mirror does not transmit light.

59. (A) A convex lens does NOT always form a virtual image: for a distant object it forms a real inverted image (catchable on a screen), so 'always virtual' is the false claim. A plane mirror's image truly is virtual and same-size. A convex mirror's image truly is always virtual and diminished. A concave mirror truly forms a real inverted image of a distant object. So the convex-lens 'always virtual' statement is the lone falsehood.

60. (B) A broad source like a tube light is made of many points spread along its length; at the halo a point on the wall has the marble blocking only some of those points while the rest of the tube is still in view, so it receives partial light and looks grey, with full darkness only where the whole tube is hidden. A tiny torch bulb is nearly a point source, so it hides all-or-nothing and gives a crisp edge. The marble does not bend light around itself, the wall does not colour the light, and light does not slow and smear past a solid object — the halo is simply partial illumination from an extended source.